Darren Biskup

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EDUCATION

Columbia University — New York, NY
M.S./Ph.D. Mechanical Engineering (Robotics) — Aug 2024 - Present
Columbia-Dream Sports AI Ph.D. Fellowship — GPA: 4.11/4.33

University of Illinois, Urbana-Champaign — Urbana, IL
B.S. Mechanical Engineering, Computer Science Minor — May 2024
High Honors, James Scholar Honors Program, Dean's List — GPA: 3.95

SKILLS

Programming & Systems: C++17, Python, C#, Linux, CMake, Git, Real-Time Programming
Controls & Estimation: Optimal Control, MPC, Trajectory Optimization, State Estimation (EKF/UKF), OSQP
Simulation & Learning: MuJoCo, PyTorch, JAX, OpenCV, Numerical Optimization, System Identification
Hardware & Integration: CAN/CAN-FD, BLDC Motor Control, Force/Torque Sensing, Pneumatics

EXPERIENCE

Robotics And Rehabilitation (ROAR) Lab, Columbia University — Aug 2024 - Present
Graduate Research Assistant

- Developed and experimentally validated a convex model predictive controller incorporating human centroidal dynamics and ground reaction force constraints for an underactuated cable-driven robot; published in IEEE RA-L. [Paper]
- Implemented the MPC as a C#/Unity QP using ALGLIB QuickQP, reducing dynamic tracking error by 24% and mechanical work by 29% versus impedance control; [Code].
- Integrated a hierarchical real-time control stack with host-level MPC, asynchronous TCP tension-command streaming, 1-kHz load-cell feedback, and CAN-controlled EPOS motor drives, incorporating Vicon/HTC Vive motion capture and force-plate sensing.

Mehta Research Group/GazzolaLab, UIUC — Jan 2023 - May 2024
Undergraduate Research Assistant - Soft Robotics

- Built an OpenCV/scikit-image calibration pipeline to extract accurate 3D marker coordinates from a custom spatial grid for robotic-arm pose tracking.
- Developed Python control scripts synchronizing a physical joystick-actuated soft robotic arm with its *pyElastica* digital twin to evaluate sim-to-real dynamic fidelity.

Skydio — May 2022 - Aug 2022
Product Design Engineering Intern

- Designed navigation-camera mounting features for the Skydio X10 and coordinated with overseas vendors to implement design changes while meeting injection-molding and lead-time constraints.
- Collaborated with manufacturing and autonomy teams on design-for-manufacturing (DFM) and spatial hardware layout for the next-generation autonomous performance quadcopter.

Lucid Motors — May 2021 - Aug 2021
Mechanical Engineering Intern, High Voltage

- Conducted data-driven thermal and efficiency analysis on high-voltage battery pack assemblies to optimize hardware performance and range.

ROBOTICS & AUTONOMY PROJECTS

Predictive Safety Filter for MuJoCo Quadrotor Flight — *Python, JAX, MuJoCo, SciPy*

- Implemented a finite-horizon predictive safety filter for online collision avoidance of a MuJoCo quadrotor in constrained workspaces with modeled obstacles. Optimizes motor commands over forward-simulated trajectories to minimize deviation from requested inputs while maintaining collision-free motion.
- Used JAX's JIT compilation and automatic differentiation to accelerate repeated nonlinear physics rollouts and SLSQP cost and constraint evaluations.

C++ MuJoCo GRF Data-Generation Pipeline — *MuJoCo, MJPC, C++, CMake*

- Developing a C++17 MuJoCo/MJPC pipeline for humanoid motion tracking and contact-wrench data generation, with morphology randomization and reproducible simulation experiments
- Implementing quasi-random morphology and marker-layout sampling across male and female humanoid models to study tracking and contact robustness.